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42.

On the Principles of the Calculus of Forms

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Commutants thus formed may be termed total commutants, because the entire of each line is made to pass through all its possible forms of arrangement. In total commutants it is necessary that the number of lines r be even; for if taken odd, on making all the r lines to change, instead of obtaining $1 \cdot 2 \cdots n$ lines, the result obtained when all but one are made to change, it will be found that the latter will be repeated $\frac{1}{2}(1 \cdot 2 \cdots n)$ times with the sign $+$, and $\frac{1}{2}(1 \cdot 2 \cdots n)$ times with the sign $-$, so that the algebraical sum of the terms will be zero. Moreover the commutants of the species above described, besides being total, are simple, inasmuch as all the umbræ to be termed consist of single letters.

My first proposition in the application of the theory of commutants to that of forms is as follows:

Let ϕ be a function homogeneous and linear in respect to an even number r of any systems whatever of variables, as

$$x_1,\ y_1 \ldots t_1; \qquad x_2,\ y_2 \ldots t_2; \qquad x_r,\ y_r \ldots t_r.$$

Form the commutant

$$\begin{matrix} \dfrac{d}{dx_1}, & \dfrac{d}{dy_1}, & \cdots \dfrac{d}{dt_1}, \\ \dfrac{d}{dx_2}, & \dfrac{d}{dy_2}, & \cdots \dfrac{d}{dt_2}, \\ \cdots & \cdots & \cdots \\ \dfrac{d}{dx_r}, & \dfrac{d}{dy_r}, & \cdots \dfrac{d}{dt_r}. \end{matrix}$$

Let the general term of this commutant, expanded, be called

$$F_{\theta_1} \times F_{\theta_2} \times \cdots \times F_{\theta_r},$$

then is

$$\Sigma F_{\theta_1} \cdot \phi \times F_{\theta_2} \cdot \phi \times \cdots \times F_{\theta_r} \cdot \phi$$

a covariant or invariant[1] as the case may be, of ϕ.

Be it observed that the march of the substitution for the different sets of variables in the above proposition is supposed to be perfectly independent. All the systems but one may undergo linear transformation, or they may all undergo distinct and disconnected transformations at the same time, and the proposition still continue applicable. It will however evidently be no less applicable should the

[1] See below, p. 324.

march of substitution for any of the systems become cogredient or contragredient to that of any other systems.

If we suppose ϕ to be a function of an even degree r of a single system of n variables $x, y \dots t$, so that the r systems x_1, y_1, &c., x_2, y_2, &c. $\dots x_r, y_r$, &c. become identical, we can at once infer from the above scheme the existence and mode of forming an invariant to ϕ of the order n. This last appears p. 308

for the case $n = 2$, and ought, for all other values of n, to have been known[2] to the author of the immortal discovery of invariants, termed by him hyperdeterminants, in the sense which, according to the nomenclature here adopted, would be conveyed by the term hyperdiscriminants.

Before proceeding to discuss the theory of compound total commutants, or enlarging upon that of partial commutants, I shall make an interesting application of the preceding general proposition to the discovery of Aronhold's S and T, the two invariants respectively of the fourth and sixth orders appertaining to a homogeneous cubic function (say F) of three variables x, y, z. These may be termed respectively H_4 and H_6. As to H_6 a theoretically possible but eminently prolix and ungraceful method immediately presents itself, namely to take $F^2 = G$, and after forming the commutant with six lines,

$$\begin{array}{ccc}
\frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\
\frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\
\frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\
\frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\
\frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\
\frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz},
\end{array}$$

to operate with the 6^5 ternary products of which this is made up upon G: the result being an invariant of G, will be so to F, and being of the third degree in

[2] That this was not known explicitly to and should have escaped the penetration of the sagacious author of the theory, and those who had studied his papers, must be attributed to the imperfection of the notation heretofore employed for denoting the coefficients of a homogeneous polynomial function. The umbral method of denoting such a function ϕ of the degree r under the form of $(ax + by + \cdots + cz)^r$, which is equivalent to, but a more compendious and independent mode of mentally conceiving and handling the representation

$$\left(x\frac{d}{dx} + y\frac{d}{dy} + \cdots + z\frac{d}{dz}\right)^r \phi,$$

exhibits the true internal constitution of such functions, and necessarily leads to the discovery of their essential properties and attributes.

respect to the coefficients of G, will be of the sixth in respect to those of F. It will evidently therefore be H_6, or at least a numerical multiple of H_6, the form of which, inasmuch as the only other invariant is H_4, we know inform to be unique. But the general theorem affords another and probably the p. 309

most practically compendious[3] solution as regards H_6, of which the question admits.

Let G[4] represent the mixed concomitant to F formed by the bordered determinant

$$\begin{vmatrix} \frac{d^2F}{dx^2} & \frac{d^2F}{dx\,dy} & \frac{d^2F}{dx\,dz} & \xi \\ \frac{d^2F}{dy\,dx} & \frac{d^2F}{dy^2} & \frac{d^2F}{dy\,dz} & \eta \\ \frac{d^2F}{dz\,dx} & \frac{d^2F}{dz\,dy} & \frac{d^2F}{dz^2} & \zeta \\ \xi & \eta & \zeta & 0 \end{vmatrix}.$$

G is a function of the second order as to x, y, z, and of the like order in respect to ξ, η, ζ, which two systems will be respectively cogredient and contragredient in respect to the x, y, z system in F. In other words, which is all we need to look to, G is a concomitant of F, and so also will be

$$G + \lambda(x\xi + y\eta + z\zeta)^2,$$

which may be termed H. Form now the commutant

$$\begin{matrix} \frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\ \frac{d}{dx}, & \frac{d}{dy}, & \frac{d}{dz}, \\ \frac{d}{d\xi}, & \frac{d}{d\eta}, & \frac{d}{d\zeta}, \\ \frac{d}{d\xi}, & \frac{d}{d\eta}, & \frac{d}{d\zeta}, \end{matrix}$$

[3]Having since this was printed been favoured with a view of some of the proof-sheets of Mr Salmon's most valuable Second Part of his *System of Analytical Geometry* (about to appear, and which is calculated, in my opinion, to awaken a higher idea of and excite a new taste for geometrical researches in this country), I find that I am mistaken in this point; the less symmetrical method operated with by Mr Salmon being decidedly of the shortest for practically obtaining S and T in the general case. Symmetry, like the grace of an eastern robe, has not unfrequently to be purchased at the expense of some sacrifice of freedom and rapidity of action.

[4]G is the mixed concomitant to the given cubic function, which is halfway (so to speak) between it and its polar reciprocal. In fact, when the operation is repeated upon G, which was executed upon the given function to obtain G (that is, when we border the Hessian of G in respect to x, y, z, vertically and horizontally with the column and line ξ, η, ζ) the determinant thereby represented becomes the polar reciprocal to the given function.

this being applied to H will give an invariant (the fact that the march of the substitutions for the systems $x, y, z; \xi, \eta, \zeta$ is contrary, being completely immaterial to the applicability of the general theorem above given); p. 310

the commutant so formed will be a cubic function of λ, in which the coefficient of λ^3 is a numerical quantity, that of λ^2 is zero, that of λ is H_4 and the constant term is H_6.

Thus for example let $F = x^3 + y^3 + z^3 + 6mxyz$, then

$$G = \begin{vmatrix} x & mz & my & \xi \\ mz & y & mx & \eta \\ my & mx & z & \zeta \\ \xi & \eta & \zeta & 0 \end{vmatrix}$$

and therefore

$$H = \Sigma\{(\lambda - m^2)x^2\xi^2 + (\lambda + m^2)2yz\eta\zeta + yz\xi^2 - 2mx^2\eta\zeta\},$$

the Σ implying the sum of similar terms with reference to the interchanges between $x, \xi;\ y, \eta;\ z, \zeta$.

In developing the commutant above, the first line may be kept in a fixed position; for the sake of brevity, $(x), (y), (z); (\xi), (\eta), (\zeta)$ may be written in the place of

$$\frac{d}{dx}, \quad \frac{d}{dy}, \quad \frac{d}{dz}; \qquad \frac{d}{d\xi}, \quad \frac{d}{d\eta}, \quad \frac{d}{d\zeta},$$

and it will readily be seen that the only effective arrangements will be as underwritten:

$$\begin{array}{llllll}
(x)(y)(z) & (x)(y)(z) & (x)(y)(z) & & & \\
(x)(y)(z) & (x)(y)(z) & (x)(y)(z) & & & \\
(\xi)(\eta)(\zeta) & (\eta)(\zeta)(\xi) & (\zeta)(\xi)(\eta) & & & \\
(\xi)(\eta)(\zeta) & (\zeta)(\xi)(\eta) & (\eta)(\zeta)(\xi) & & & \\[1ex]
(x)(y)(z) & (x)(y)(z) & (x)(y)(z) & (x)(y)(z) & (x)(y)(z) & (x)(y)(z) \\
(x)(z)(y) & (x)(z)(y) & (\xi)(\eta)(\zeta) & (z)(y)(x) & (y)(x)(z) & (y)(x)(z) \\
(\xi)(\eta)(\zeta) & (\xi)(\zeta)(\eta) & (\xi)(\eta)(\zeta) & (\zeta)(\eta)(\xi) & (\xi)(\eta)(\zeta) & (\eta)(\xi)(\zeta) \\
(\xi)(\zeta)(\eta) & (\xi)(\eta)(\zeta) & (\xi)(\eta)(\zeta) & (\xi)(\eta)(\zeta) & (\eta)(\xi)(\zeta) & (\xi)(\eta)(\zeta) \\[1ex]
(x)(y)(z) & (x)(y)(z) & (x)(y)(z) & (x)(y)(z) & (x)(y)(z) & (x)(y)(z) \\
(y)(z)(x) & (z)(x)(y) & (y)(z)(x) & (y)(z)(x) & (z)(x)(y) & (z)(x)(y) \\
(\xi)(\eta)(\zeta) & (\eta)(\zeta)(\xi) & (\zeta)(\eta)(\xi) & (\eta)(\zeta)(\xi) & (\xi)(\eta)(\zeta) & (\zeta)(\xi)(\eta) \\
(\xi)(\zeta)(\eta) & (\eta)(\xi)(\zeta) & (\eta)(\zeta)(\xi) & (\xi)(\eta)(\zeta) & (\zeta)(\eta)(\xi) & (\xi)(\eta)(\zeta).
\end{array}$$

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The signs of the four lines in each of these arrangements are two alike, and two contrary to the signs of the correspondent lines in the first arrangement; hence the effective sign is the same for all, and the result, after rejecting from each term the common factor -16, is seen, from inspection, to be

$$4(\lambda - m^2)^3 - 8m^3 + 6(\lambda - m^2)(\lambda + m^2)^2 - 12m(\lambda + m^2)^2 + 2(\lambda + m^2)^3 + 1,$$

which is equal to

$$12\lambda^3 + 0 \cdot \lambda^2 - 12(m - m^4)\lambda + 1 - 20m^3 - 8m^6;$$

here the coefficients $m-m^4$ and $1-20m^3-8m^6$ are the two invariants (Aronhold's S and T) for the canonical form operated upon; and it will be observed that

$$(1-20m^3-8m^6)^2+64(m-m^4)^3=(1+8m^3)^3,$$

which is easily proved to be the discriminant of

$$x^3+y^3+z^3+6mxyz.$$

It may however be observed, that this is not the discriminant of the function in λ just found, as reasons of analogy[5] might have suggested it probably would be: in order that this might be the case, the coefficient of λ^3 should be 4 instead of 12, and of λ, $m-m^4$ instead of m^4-m. There is ground for supposing that another function of λ may be found by a different method, in which this relation will take effect.

The theorem above given for simple total commutants admits of an interesting application to the general case of a function F of the nth degree, in respect to each of two independent systems of two variables $x, y; \xi, \eta$. Let F be symbolically represented by $(ax+by)^n(\alpha\xi+\beta\eta)^n$, so that $a^n\alpha^n$ represents the coefficient of $x^n\xi^n$, $na^{n-1}b\alpha^n$ of $x^{n-1}y\xi^n$, &c. &c.; then the commutant

$$\begin{array}{ccc}
a, & b, & (1) \\
a, & b, & (2) \\
\cdots & \cdots & \\
a, & b, & (n) \\
\alpha, & \beta, & (1) \\
\alpha, & \beta, & (2) \\
\cdots & \cdots & \\
\alpha, & \beta, & (n)
\end{array}$$

will represent a quadratic invariant of F, which will contain $(n+1)^2$ coefficients. By expanding this commutant we obtain a general expression for the invariant under a very interesting form. p. 312

I now proceed to give the general theorem for compound total commutants as applicable to the discovery of invariants.

Let there be a function of m disconnected classes of systems of variables; let the systems in the same class be supposed all distinct but cogredient with one another. The function is supposed to be linear in respect to each system in each class, and the number of systems is the same for all the classes, and the number

[5]The biquadratic function of x, y having only one parameter, and therefore two invariants, its theory possesses striking analogies to the theory of the cubic function of three letters. The function in λ which gives these invariants for the first-named function, according to the method given in the first section, has the same discriminant as the function itself.

of variables the same in each system. This function may then be represented symbolically under the form

$$\begin{aligned}
&({}^1a_1\,{}^1x_1 + {}^1b_1\,{}^1y_1 + \cdots + {}^1l_1\,{}^1t_1)({}^1a_2\,{}^1x_2 + {}^1b_2\,{}^1y_2 + \cdots + {}^1l_2\,{}^1t_2)\\
&\quad\cdots({}^1a_n\,{}^1x_n + {}^1b_n\,{}^1y_n + \cdots + {}^1l_n\,{}^1t_n)\\
&\times({}^2a_1\,{}^2x_1 + {}^2b_1\,{}^2y_1 + \cdots + {}^2l_1\,{}^2t_1)({}^2a_2\,{}^2x_2 + {}^2b_2\,{}^2y_2 + \cdots + {}^2l_2\,{}^2t_2)\\
&\quad\cdots({}^2a_n\,{}^2x_n + {}^2b_n\,{}^2y_n + \cdots + {}^2l_n\,{}^2t_n)\times \&c.\\
&\times({}^pa_1\,{}^px_1 + {}^pb_1\,{}^py_1 + \cdots + {}^pl_1\,{}^pt_1)({}^pa_2\,{}^px_2 + {}^pb_2\,{}^py_2 + \cdots + {}^pl_2\,{}^pt_2)\\
&\quad\cdots({}^pa_n\,{}^px_n + {}^pb_n\,{}^py_n + \cdots + {}^pl_n\,{}^pt_n).
\end{aligned}$$

In this expression the $x, y \dots t$'s are all real, but the $a, b \dots l$'s all umbral; in fact, ${}^fa_g, {}^fb_g$, &c. may be understood to denote

$$\frac{d}{d\,{}^fx_g}, \qquad \frac{d}{d\,{}^fy_g}, \quad \&c.$$

The n systems of variables in each of the sets above written are supposed to be cogredient *inter se*.

Take the symbolical product of the first set, first making *for the moment*

$${}^1x_1 = {}^1x_2 = \cdots {}^1x_n = x, \quad \&c.\ \&c., \qquad {}^1t_1 = {}^1t_2 = \cdots {}^1t_n = t;$$

and let the coefficients of the several terms

$$x^n, \quad x^{n-1}y \dots \&c.,$$

be called

$${}^1A_1, \quad {}^1A_2, \dots {}^1A_\mu,$$

where μ is the number of terms contained in a homogeneous function of the nth degree of the m variables $x, y \dots t$. In like manner proceed with each of the lines, and then write down the commutant

$$\begin{matrix}
{}^1A_1, & {}^1A_2, & \cdots & {}^1A_\mu,\\
{}^2A_1, & {}^2A_2, & \cdots & {}^2A_\mu,\\
\cdots & \cdots & & \cdots\\
{}^pA_1, & {}^pA_2, & \cdots & {}^pA_\mu.
\end{matrix}$$

This commutant is an invariant of F: it will of course be remembered that, unless p is even, the commutant vanishes. p. 313

Thus, for example, take two sets of two systems of two variables: in all four systems,

$$x, y; \quad \xi, \eta: \quad p, q; \quad \phi, \psi,$$

each couple of systems on either side of the colon (:) being cogredient *inter se*: and let F be symbolically represented by

$$(ax + by)(\alpha\xi + \beta\eta)(lp + mq)(\lambda\phi + \mu\psi);$$

then the invariant given by the theorem will be the commutant

$$\begin{array}{lll} a\alpha; & a\beta+\alpha b; & b\beta, \\ l\lambda; & l\mu+\lambda m; & m\mu. \end{array}$$

The six positions of this are as below written (the first three being positive and the second three negative)

$$\begin{array}{lll} a\alpha;\ a\beta+\alpha b;\ b\beta, & a\alpha;\ a\beta+\alpha b;\ b\beta, & a\alpha;\ a\beta+\alpha b;\ b\beta, \\ l\lambda;\ l\mu+\lambda m;\ m\mu, & l\mu+\lambda m;\ m\mu;\ l\lambda, & m\mu;\ l\lambda;\ l\mu+\lambda m, \end{array}$$

$$\begin{array}{lll} a\alpha;\ a\beta+\alpha b;\ b\beta, & a\alpha;\ a\beta+\alpha b;\ b\beta, & a\alpha;\ a\beta+\alpha b;\ b\beta, \\ l\mu+\lambda m;\ l\lambda;\ m\mu, & l\lambda;\ m\mu;\ l\mu+\lambda m, & m\mu;\ l\mu+\lambda m;\ l\lambda. \end{array}$$

If we write F under its explicit form,

$$\begin{aligned} & Ax\xi p\phi + Bx\xi p\psi + Cx\xi q\phi + Dx\xi q\psi \\ & + A'x\eta p\phi + B'x\eta p\psi + C'x\eta q\phi + D'x\eta q\psi \\ & + A''y\xi p\phi + B''y\xi p\psi + C''y\xi q\phi + D''y\xi q\psi \\ & + A'''y\eta p\phi + B'''y\eta p\psi + C'''y\eta q\phi + D'''y\eta q\psi, \end{aligned}$$

we have identically the relations following,

$$\begin{array}{llll} a\alpha l\lambda = A, & a\alpha l\mu = B, & a\alpha m\lambda = C, & a\alpha m\mu = D, \\ a\beta l\lambda = A', & a\beta l\mu = B', & a\beta m\lambda = C', & a\beta m\mu = D', \\ b\alpha l\lambda = A'', & b\alpha l\mu = B'', & b\alpha m\lambda = C'', & b\alpha m\mu = D'', \\ b\beta l\lambda = A''', & b\beta l\mu = B''', & b\beta m\lambda = C''', & b\beta m\mu = D'''. \end{array}$$

and the commutant expanded becomes

$$\begin{aligned} & A(B'+C''+C'+B'')D''' + (B+C)(D'+D'')A''' \\ & \quad + D(A'+A'')(B'''+C''') - (B+C)(A'+A'')D''' \\ & \quad - A(D'+D'')(B'''+C''') - D(B'+C''+C'+B'')A'''. \end{aligned}$$

In the foregoing the x's in the several lines were *for the moment* taken identical, in order the more easily to explain the law of formation of the p. 314

quantities A. But suppose that they become actually identical for the same line. F then becomes a function of the nth degree in respect to each of p systems of variables, and may be represented symbolically under the form

$$({}^1a\,{}^1x + {}^1b\,{}^1y + \cdots + {}^1l\,{}^1t)^n \times ({}^2a\,{}^2x + {}^2b\,{}^2y + \cdots + {}^2l\,{}^2t)^n \cdots \times ({}^pa\,{}^px + {}^pb\,{}^py + \cdots + {}^pl\,{}^pt)^n.$$

We may still further limit the generality of the theorem by supposing

$${}^1x = {}^2x = \cdots {}^px = x, \qquad {}^1y = {}^2y = \cdots {}^py = y, \qquad \cdots, \qquad {}^1t = {}^2t = \cdots {}^pt = t;$$

F then becomes

$$(ax + by + \cdots + lt)^{np}.$$

Accordingly, as many different factors as can be found contained an even number of times in the exponent of the function, so many invariants can be formed immediately from a function of any number of variables m by the method of total commutation.

If one of these factors be called n, the commutant corresponding thereto will be of the order

$$\frac{(n+1)(n+2)\cdots(n+m-1)}{1\cdot 2\cdots(m-1)}$$

in respect to the coefficients. Thus take $m = 2$, so that

$$F = (ax + by)^{np}.$$

The general form of such a commutant will be found by taking $A_1, A_2 \ldots A_{n+1}$ the coefficients of the several combinations of x, y in $(ax + by)^n$, from which the numerical coefficients n, $\frac{1}{2}n(n-1)$, &c. may be rejected, as only introducing a numerical factor into the result; the commutant will therefore be expressed by means of the form

$$\begin{array}{lllll} a^n; & a^{n-1}b; & a^{n-2}b^2 \ldots; & b^n, & (1) \\ a^n; & a^{n-1}b; & a^{n-2}b^2 \ldots; & b^n, & (2) \\ & & \cdots\cdots\cdots\cdots\cdots & & \\ a^n; & a^{n-1}b; & a^{n-2}b^2 \ldots; & b^n. & (p) \end{array}$$

If $p = 2$, the compound commutant

$$\begin{array}{llll} a^n; & a^{n-1}b; & \cdots; & b^n, \\ a^n; & a^{n-1}b; & \cdots; & b^n, \end{array}$$

will easily be seen to be only another form for the catalecticant of $(ax + by)^{2n}$. p. 315
Thus, let $n = 2$,

$$(ax + by)^4 = Ax^4 + 4Bx^3y + 6Cx^2y^2 + 4Dxy^3 + Ey^4;$$

so that

$$a^4 = A, \qquad a^3b = B, \qquad a^2b^2 = C, \qquad ab^3 = D, \qquad b^4 = E.$$

The commutant (which is of the form of the matrix to an ordinary determinant, with the exception that the umbræ enter compoundly instead of simply into the several terms separated by the marks of punctuation), will be

$$\begin{array}{lll} a^2; & ab; & b^2, \\ a^2; & ab; & b^2 : \end{array}$$

this, written in the six forms

$$\left.\begin{matrix} a^2; & ab; & b^2 \\ a^2; & ab; & b^2 \end{matrix}\right\} \quad \left.\begin{matrix} a^2; & ab; & b^2 \\ a^2; & b^2; & ab \end{matrix}\right\} \quad \left.\begin{matrix} a^2; & ab; & b^2 \\ ab; & a^2; & b^2 \end{matrix}\right\}$$
$$\left.\begin{matrix} a^2; & ab; & b^2 \\ b^2; & ab; & a^2 \end{matrix}\right\} \quad \left.\begin{matrix} a^2; & ab; & b^2 \\ ab; & b^2; & a^2 \end{matrix}\right\} \quad \left.\begin{matrix} a^2; & ab; & b^2 \\ b^2; & a^2; & ab \end{matrix}\right\}$$

gives the expression

$$a^4 \times a^2b^2 \times b^4 - a^4 \times (ab^3)^2 - b^4 \times (a^3b)^2 - (a^2b^2)^3 + 2a^3b \times ab^3 \times a^2b^2;$$

that is

$$ACE - AD^2 - EB^2 - C^3 + 2BCD.$$

One important observation may here be made of a fact which otherwise might easily escape attention, which is, that commutants, where the same terms simple or compound are found in all or several of the lines, in general give rise to products, some of them equal and with the same sign, and others equal but with the contrary sign.

This last phenomenon does not manifest itself in commutants appertaining to functions of two variables of the two particular and different species which first and most naturally present themselves, namely where there are only two lines or only two columns[6]—I believe that it displays itself in every other case of commutantives to functions of two variables. Thus it is that algebraical expressions derived from given functions disguise their symmetry; to make which come to light it becomes necessary to add terms of contrary sign to such expressions. As an example, the reader is invited to develope the cubic invariant of a function of x and y, symbolically expressed by $(ax + by)^8$, where

$$a^8 = A, \qquad a^7b = B \ldots ab^7 = H, \qquad b^8 = I,$$

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by means of the commutant

$$\begin{matrix} a^2, & ab, & b^2, \\ a^2, & ab, & b^2, \\ a^2, & ab, & b^2, \\ a^2, & ab, & b^2. \end{matrix}$$ [7]

Suppose F to be the general even-degreed function of two variables of the degree $2np$.

[6]These commutants give respectively the quadrinvariant and the catalecticant, the former of which alone was formerly recognised by Mr Cayley as a commutant.

[7]See p. 346 below. The number of terms resulting from the independent permutation of each of the 3 linear lines is 6^3, that is 216; but the actual result is (using small letters instead

Let

$$H = \left(\xi \frac{d}{dy} - \eta \frac{d}{dx}\right)^{np} F + \lambda(x\xi + y\eta)^{np},$$

and express H umbrally under the form

$$(ax + by)^{np}(\alpha\xi + \beta\eta)^{np}.$$

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of large) $P - Q$, where

$$P = aei + 3ag^2 + 12beh + 3c^2i + 24cf^2 + 24d^2g + 15e^3,$$
$$Q = 4afh + 4bid + 8bgf + 22ceg + 8chd + 36def,$$

so that the effective number of permutations is only 164. The difference between this and 216 divided by 216 may be termed the Index of Demolition, which we see in this case is $\frac{52}{216}$ or $\frac{13}{54}$; that is, somewhat less than $\frac{1}{4}$. For the cubic invariant of the function of the fourth degree this index is zero, all the permutations being effective. If we take the cubic invariant of the function $ax^{12} + 12bx^{11}y + 66cx^{10}y^2 + \&c. + my^{12}$ under the form $P - Q$, we shall find

$$\begin{aligned} P &= 6ahl + 10ajj + 60fm + 54bhk + 54cfl + 156cii + 10ddm + 430djj \\ &\quad + 155eek + 520ehh + 520ffl + 280ggg, \\ Q &= agm + 15aik + 30bgl + 50bij + 15cem + 4cgk + 150chj + 30del + 210dfk \\ &\quad + 250dhi + 230efj + 555egl + 660fgh. \end{aligned}$$

The number of terms in P and Q is of course the same, and will be found to be 2200 for each; so that out of the 6^5, that is 7776 permutations of the 5 lower rows, only 4400 are effective, and the index of demolition becomes $\frac{3376}{7776}$, that is $\frac{211}{486}$, or rather greater than $\frac{5}{12}$. The Index of Demolition thus goes on constantly increasing as the degree of the function rises; probably (?) it converges either towards $\frac{1}{2}$ or else towards unity. In arranging the terms it will be found most convenient to adopt, as I have done above, the dictionary method of sequence. The computations are greatly facilitated by the circumstance of the effect of any arrangement of each of the 5 lower lines not being altered when these lines are permuted with one another; this gives rise to the subdivision of the 7776 permutations into groups as follows: 6 of 120 identical terms, 60 of 60, 36 of 20, 60 of 30, 24 of 20, 30 of 10, 30 of 5, and 6 of 1. So that the total number of permutational arrangements to be constructed is only 252. Other methods of abridging the labour will readily suggest themselves to the practical computer. The total number of the groups of terms is of course always known *à priori*, and, for instance, in the case before us, must be equal to the number of ways in which $\frac{1}{2}(12 \times 3)$, that is the number 18, can be divided into 3 parts, none of which is to exceed the number 12, that is 25; for the cubic invariant of the function of the eighth degree of two variables it is the number of ways in which 12 can be divided into 3 parts, of which none shall exceed 8, and so forth, zeros being always understood to be admissible; and of course in general for an invariant of the order r to a function of the degree n of i variables, the number of distinct terms is in general the number of ways in which $\frac{nr}{i}$ can be divided into r parts, of which none shall exceed n, subject however always to the possibility in particular cases of a diminution in consequence of some of the groups assuming zero for their coefficient.

The commutant

$$\begin{array}{llcll}
a^n, & a^{n-1}b & \cdots & b^n, & (1) \\
a^n, & a^{n-1}b & \cdots & b^n, & (2) \\
 & \multicolumn{3}{c}{\dots\dots\dots\dots} & \\
a^n, & a^{n-1}b & \cdots & b^n, & (p) \\
\alpha^n, & \alpha^{n-1}\beta & \cdots & \beta^n, & (1) \\
\alpha^n, & \alpha^{n-1}\beta & \cdots & \beta^n, & (2) \\
 & \multicolumn{3}{c}{\dots\dots\dots\dots} & \\
\alpha^n, & \alpha^{n-1}\beta & \cdots & \beta^n, & (p)
\end{array}$$

will be a function of λ, and all the several coefficients will be invariants of F[8].

When $p = 1$ we obtain the Λ given in the preceding section, and originally published by me in the *Philosophical Magazine* for the month of November, 1851. The Λ obtained on this supposition has for its coefficients a series of independent invariants, commencing with the catalecticant and closing with the quadratic invariant. When p has any other value, we observe a similar series commencing with a commutantive invariant of a lower order than the catalecticant, but always closing with the quadratic invariant. Thus, for example, when $2np = 8$, we may obtain by the preceding theorem three different quadratic functions; one giving the invariants of the orders 5, 4, 3, 2, the second those of the orders 3, 2, the third the invariant of the order 2.

In this case the invariants of the same order given by the different Λ's are the same to numerical factors *près*. Whether this is always necessarily the case is a point reserved for further examination.

The commutants applied in the preceding theorems have been called by me total commutants, because the total of each line of umbræ is permuted in every possible manner. If the lines be divided into segments, and the permutation be local for each segment instead of extending itself over the whole line, we then arrive at the notion of partial commutants, to which I have also (in concert with Mr Cayley) given the distinctive name of Intermutants. In order to find the invariants of functions of odd degrees, the theory of total commutants requires the process of commutation to be applied, not immediately to the coefficients of the proposed function, but to some derived concomitant form. I became early sensible of this imperfection, and stated to the friend above named, to whom I had previously p. 318

imparted my general method of total commutation, my conviction of the existence of a qualified or restricted method of permutation, whereby the invariants of the cubic function, for instance, of two and of three letters would admit,

[8]By substituting the symbols $\frac{d}{dx}$, $\frac{d}{dy}$, &c. in place of the umbræ a, b, &c., the theorem is easily stated for covariants generally. But in applying the commutantive method to obtain covariants, or rather in the statement of the results flowing from each application, it is never necessary to go beyond the case of invariants, because the commutantive covariants of any given homogeneous function are always identical with commutantive invariants of emanants of the same function.

without the aid of a derived form, of being represented. Many months ago, when I was engaged in this important research, and had made some considerable steps towards the representation of the invariant, that is, the discriminant of the cubic function of x and y, under the form of a single permutant, I was surprised by a note from the friend above alluded to, announcing that he had succeeded in fixing the form of the permutant of which I was at that moment in search. It is with no intention of complaining of this interference on the part of one to whose example and conversation I feel so deeply indebted, (and the undisputed author of the theory of Invariants,) that I may be permitted to say that, independent of the intervention of this communication, I must inevitably have succeeded in shaping my method so as to furnish the form in question; and that with greater certainty, after my theory of commutants had furnished me with the precedent of permutable forms giving rise to terms identical in value but affected with contrary signs. As I have understood that Mr Cayley is likely to develop this part of the subject in the present number of the *Journal*, it will be the less necessary for me to enter at any length into the theory of partial commutants on the present occasion.

The method of partial commutation is a simple but most important corollary from that of total commutation hereinbefore explained. To fix the ideas, conceive a class of p cogredient systems, and that there are qr such classes perfectly independent. Proceed to divide these qr classes in any manner whatever into r sets, each containing q classes; and form the symbol of the total commutant corresponding to each such set. Now let these commutants be placed side by side against one another, and transpose the terms in each compound line thus formed once for all, but in any arbitrary manner. Then permute in every possible way all those symbols in each line, *inter se*, which belong to the same class, and operate with the symbols thus produced by reading off the vertical columns and attending to the rule of the + and − signs, as in the case of a total commutant; the result will be a commutant of the form operated upon. For instance, let $p = 1,\ q = 3,\ r = 2$, and let the number of variables in each system be 2. Form the commutant operators

$$\begin{array}{cc|cc}
\dfrac{d}{dx}, & \dfrac{d}{dy}, & \dfrac{d}{d\xi}, & \dfrac{d}{d\eta}, \\
\dfrac{d}{dp}, & \dfrac{d}{dt}, & \dfrac{d}{d\phi}, & \dfrac{d}{d\theta}, \\
\dfrac{d}{dr}, & \dfrac{d}{ds}, & \dfrac{d}{d\rho}, & \dfrac{d}{d\sigma}.
\end{array}$$

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Interchange *in any manner* but *once for all* the symbols in each line, as thus:

$$\begin{array}{cccc} \dfrac{d}{dx}, & \dfrac{d}{dy}, & \dfrac{d}{d\xi}, & \dfrac{d}{d\eta}, \\ \dfrac{d}{d\phi}, & \dfrac{d}{dp}, & \dfrac{d}{dt}, & \dfrac{d}{d\theta}, \\ \dfrac{d}{ds}, & \dfrac{d}{d\rho}, & \dfrac{d}{dr}, & \dfrac{d}{d\sigma}. \end{array}$$

Now permute, *inter se*, the variables of each system, as

$$\frac{d}{dx}, \frac{d}{dy}; \qquad \frac{d}{dp}, \frac{d}{dt}, \quad \&c.;$$

the total number of the operative forms resulting will be $(1\cdot 2)^6$, and the sum of the $(1\cdot 2)^6$ quantities, half positive and half negative, formed after the type of

$$\Sigma\left\{\frac{d}{dx}\frac{d}{d\phi}\frac{d}{ds}U \times \frac{d}{dy}\frac{d}{dp}\frac{d}{d\rho}U \times \frac{d}{d\xi}\frac{d}{d\theta}\frac{d}{dr}U \times \frac{d}{d\eta}\frac{d}{dt}\frac{d}{d\sigma}U\right\},$$

U being supposed to be a function homogeneous in

$$x, y; \quad \xi, \eta; \quad p, t; \quad \phi, \theta; \quad r, s; \quad \rho, \sigma,$$

will be a covariant of U.

The proof of the truth of this proposition is contained in what is shown in the Notes of the Appendix for total commutants, it being only necessary to make the systems which are independent vary consecutively, and then apply the inference to the supposition of their varying simultaneously.

It may be extended to the more general supposition of classes of an unequal number of cogredient systems of unequal numbers of variables in each, the only condition apparently required being that the number of distinct terms shall be the same in each line of the final commutantive operator. The important remark to be made is, that in applying this theorem there is nothing to prevent any of the systems being made identical; or, in other words, a given function of one system of variables may be regarded as a function of as many different, although coincident, sets as we may choose to suppose. Thus, suppose

$$U = Ax^2 + 2Bxy + Cy^2;$$

we may take the partial commutant formed of the two total commutant operators

$$\begin{array}{cc} \dfrac{d}{dx}, & \dfrac{d}{dy}, \\ \dfrac{d}{dx}, & \dfrac{d}{dy}, \end{array}$$

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combined with itself. If we write them in the same order,

$$\begin{array}{cccc}
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy},
\end{array}$$

(where I use the dots and dashes to distinguish those in the same line which are considered as belonging to the same class, and therefore as permutable, *inter se*), we shall evidently obtain $4\{AC - B^2\}^2$; if we commence with a permutation, so as to have the form of operation

$$\begin{array}{cccc}
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{d'}{dy}, & \dfrac{d'}{dx}, & \dfrac{\dot{d}}{dy},
\end{array}$$

it will be found that we obtain $2\{AC - B^2\}^2$.

Again, suppose that we have

$$U = Ax^3 + 3Bx^2y + 3Cxy^2 + Dy^3.$$

If we write

$$\begin{array}{cccc}
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy},
\end{array}$$

the value of the commutant would come out zero; but if we make a permutation, and write

$$\begin{array}{cccc}
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{d'}{dy}, & \dfrac{d'}{dx}, & \dfrac{\dot{d}}{dy}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{\dot{d}}{dx}, & \dfrac{d'}{dy},
\end{array}$$

the operation indicated by the above performed upon U, will give a multiple of the discriminant of U. p. 321

In like manner we may represent Aronhold's Sextic Invariant of the form

$(x, y, z)^3$ by means of the partial commutant

$$
\begin{array}{cccccc}
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{\dot{d}}{dz}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, & \dfrac{d'}{dz}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{\dot{d}}{dz}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, & \dfrac{d'}{dz}, \\
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{\dot{d}}{dz}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, & \dfrac{d'}{dz}.
\end{array}
$$

If we make

$$V = \left(\xi' \frac{d}{d\xi} + \eta' \frac{d}{d\eta} + \zeta' \frac{d}{d\zeta}\right) \left(\xi \frac{d}{dx} + \eta \frac{d}{dy} + \zeta \frac{d}{dz}\right)^2 (x, y, z)^3,$$

and use H to signify the determinant

$$\begin{vmatrix} x & y & z \\ \xi & \eta & \zeta \\ \xi' & \eta' & \zeta' \end{vmatrix},$$

which is evidently an universal triple covariant, and make

$$W = V + \lambda H,$$

and apply to W the partial commutantive symbol

$$
\begin{array}{cccccc}
\dfrac{\dot{d}}{dx}, & \dfrac{\dot{d}}{dy}, & \dfrac{\dot{d}}{dz}, & \dfrac{d'}{dx}, & \dfrac{d'}{dy}, & \dfrac{d'}{dz}, \\
\dfrac{d}{d\xi}, & \dfrac{d}{d\eta}, & \dfrac{d}{d\zeta}, & \dfrac{d'}{d\xi}, & \dfrac{d'}{d\eta}, & \dfrac{d'}{d\zeta}, \\
\dfrac{d}{d\xi'}, & \dfrac{d}{d\eta'}, & \dfrac{d}{d\zeta'}, & \dfrac{d'}{d\xi'}, & \dfrac{d'}{d\eta'}, & \dfrac{d'}{d\zeta'},
\end{array}
$$

we shall obtain a function of λ of which all the odd powers and the second power will disappear, and such that the coefficients of λ^2 and the constant term will be Aronhold's S and T, and the discriminant of the entire function in respect to λ^2 (if not for the distribution assigned to the dots and dashes in the foregoing, at least for some other distribution) may not improbably be the discriminant of the given function $(x, y, z)^3$. p. 322

Notes in Appendix.

(1) [p. 295 above.] More generally, in as many ways as the number n can be divided into parts, in so many ways can a given function of one set of variables be as it were unravelled so as to furnish concomitant forms.

For instance, the form $ax^3 + 3bx^2y + 3cxy^2 + dy^3$ has for a concomitant

$$aux + buy + bvx + cvy + cwx + dwy,$$

where u, v, w are cogredient with $x^2, 2xy, y^2$; and also

$$auu'x + buuy + buv'x + bvu'x + cvv'x + cvu'y + cuv'y + dvv'y,$$

where u, v; u', v' are cogredient with each other and with x and y; and the proposition in the text may be best derived from this more general theorem by dividing the index into equal parts, forming as many systems as there are such parts, and then identifying the systems so formed.

(2) [p. 297 above.] The following additional example will illustrate the power of this method.

Let $\phi = (x, y, z)^4$ be the general function of the fourth degree. Form by *unravelment* the concomitant form $(u, v, w, p, q, r)^2$ (say P) where u, v, w, p, q, r are cogredient with $x^2, y^2, z^2, 2zy, 2xz, 2yx$.

Again, the universal concomitant $(x\xi + y\eta + z\zeta)^2$ will have for its concomitant

$$u\xi^2 + v\eta^2 + w\zeta^2 + p\eta\zeta + q\zeta\xi + r\xi\eta,$$

where ξ, η, ζ are contragredient to x, y, z. Now take the reciprocal polar of this last form with respect to ξ, η, ζ; that is,

$$\Sigma\left(vw - \frac{1}{4}p^2\right)x_1^2 + 2\Sigma\left(\frac{1}{4}qr - \frac{1}{2}pu\right)y_1z_1, \qquad \text{(say } G\text{)},$$

where x_1, y_1, z_1, being contragredient to ξ, η, ζ, will be cogredient with x, y, z. $P + \lambda G$ is a quadratic function of the six variables u, v, w, p, q, r, and its discriminant will give a function of λ of the sixth degree, all of whose even coefficients will be covariants of ϕ. If we replace x_1, y_1, z_1 by x, y, z, these even coefficients will be respectively (understanding that order refers to the dimensions *quoad* the coefficients of ϕ and degree to the dimensions *quoad* x, y, z) as follows: p. 323

Of order 6	degree	0,
" 5	"	2,
" 4	"	4,
" 3	"	6,
" 2	"	8,
" 1	"	10,
" 0	"	12.

The two last coefficients must evidently be identically zero. It is *possible* that some of the others may be so too: as regards the one of the third order and sixth degree, this is of the same form as, and may be identical with, the Hessian of ϕ; as regards the one of the fourth order and fourth degree, this may be ϕ

itself multiplied by the cubic invariant (which the theory of Section III. proves to exist) of ϕ. But the covariants of the fifth order and second degree, and of the second order and eighth degree, if they are not identically zero, and if the latter is not ϕ^2 (which a trial or two of some very simple cases will easily establish one way or the other) are probably irreducible forms. The existence of a correlated conic section to a curve of the fourth order, if established, would be particularly interesting, and its geometrical meaning would well deserve being elicited.

(3) [p. 303 above.] If any form (f) of the degree n be written symbolically,

$$(a_1x_1 + a_2x_2 + \cdots + a_\iota x_\iota)^n,$$

where $x_1, x_2 \ldots x_\iota$ are real but $a_1, a_2 \ldots a_\iota$ umbral, and if I_r be any invariant of the order r in respect of the real coefficients of (f), it is easily seen by reason of I_r remaining unaltered when $x_1, x_2 \ldots x_\iota$ become respectively $f_1x_1, f_2x_2 \ldots f_\iota x_\iota$, provided that $f_1, f_2 \ldots f_\iota = 1$, that each term in I_r expressed by means of the umbræ, must contain an equal number of times $a_1, a_2 \ldots a_\iota$, so that each such term will contain $\frac{nr}{\iota}$ of each of them, of course differently subdivided and grouped; hence we have the universal condition that $\frac{nr}{\iota}$ must be an integer; but this is less stringent than the actual condition, which is that $\frac{nr}{\iota}$ must be an integer of a certain form; for instance, as before observed, when $\iota = 2$, $\frac{nr}{\iota}$ must be an *even* integer.

(4) [p. 307 above.] To prove the theorem given in the text for total simple commutants it is only necessary to bear in mind that whenever two columns in any total commutant become identical, the commutant vanishes. To fix the ideas, take the commutant formed of lines similar to $\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}$, written p. 324

under one another; let there be r such lines, the total number of terms will be $(1 \cdot 2 \cdot 3)^r$: the $1 \cdot 2 \cdot 3$ positions of the line written above will correspond to $(1 \cdot 2 \cdot 3)^{r-1}$ several groupings of the remaining lines. Now when x, y, z undergo a unimodular linear substitution, $\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}$ will undergo a related substitution not coincident with that of x, y, z, but still unimodular; let x, y, z change, all the other systems remaining fixed, and suppose

$$\frac{d}{dx}, \quad \frac{d}{dy}, \quad \frac{d}{dz}$$

to become respectively

$$f\frac{d}{dx} + g\frac{d}{dy} + h\frac{d}{dz}, \qquad f'\frac{d}{dx} + g'\frac{d}{dy} + h'\frac{d}{dz}, \qquad f''\frac{d}{dx} + g''\frac{d}{dy} + h''\frac{d}{dz},$$

then each of the $(1 \cdot 2 \cdot 3)^{r-1}$ groups of the terms arising from the permutation of $\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}$ will subdivide into 27 groups, of which we may reject those in which any of the terms $\left(\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}\right)$ occurs twice or three times; accordingly there will be left only the six effective orders of permutations,

$$\left(f\frac{d}{dx},\ g'\frac{d}{dy},\ h''\frac{d}{dz}\right); \qquad \left(f\frac{d}{dx},\ h'\frac{d}{dz},\ g''\frac{d}{dy}\right); \quad \&c.$$

consequently each of the $(1 \cdot 2 \cdot 3)^{r-1}$ groups gives rise to 6 times 6 products whose sum will be

$$\begin{vmatrix} f'' & g'' & h'' \\ f' & g' & h' \\ f & g & h \end{vmatrix} \times \text{ the sum of the 6 products corresponding}$$

to the permutations of $\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}$; and therefore, the transformations being unimodular, the sum of the products corresponding to the entire $(1 \cdot 2 \cdot 3)^r$ permutations remains constant when x, y, z change. In like manner, all the systems may change one after the other, and consequently all of them at the same time without affecting the value of the commutant: and in like manner for the general case. Q.E.D.

(5) [p. 312 above.] The truth of the proposition relative to compound commutants and the mode of the demonstration will be apparent from the subjoined example.

Let the function be supposed to be

$$(ax + by)(a'x' + b'y')(\alpha\xi + \beta\eta)(\alpha'\xi' + \beta'\eta'),$$

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where $x, y;\ x', y'$ are cogredient and $\xi, \eta;\ \xi', \eta'$ cogredient; the a, b, α, β, &c. are of course mere umbræ. Now take the compound commutant

$$\begin{matrix} aa', & ab' + a'b, & bb', \\ \alpha\alpha', & \alpha\beta' + \alpha'\beta, & \beta\beta'. \end{matrix}$$

Let $x, y;\ x', y'$ undergo a linear substitution, and, accordingly,

$$\begin{matrix} a & \text{become} & fa + gb, \\ a' & \text{''} & fa' + gb', \\ b & \text{''} & ha + kb, \\ b' & \text{''} & ha' + kb', \end{matrix}$$

f, g, h, k being of course actual and not umbral; then the above commutant will be easily seen to decompose into 6 others, which will be equal to the original commutant multiplied by the determinant

$$\begin{vmatrix} f^2 & 2fg & g^2 \\ fh & fk + gh & gk \\ h^2 & 2hk & k^2 \end{vmatrix}$$

which is equal to $(fk - gh)^3$, that is $= 1$.

And so in general, which shows, as in the preceding note, that all the classes of cogredient systems may be transformed successively one after the other, and therefore simultaneously, without altering the value of the commutant.

(6) In the last May Number[9] of the *Journal*, Mr Boole, to whose modest labours the subject is perhaps at least as much indebted as to any one other writer, has given a theorem[10], (14) p. 94, the excellent idea contained in which there is no difficulty in shaping so as to render it generalizable by aid of the theory of contravariants. It may be regarded in some sort a pendant or reciprocal to the Eisenstein-Hermite theorem, presented by me under a wider aspect in the First Section of this paper. p. 326

Let $\phi(x, y \dots z)$ have any contravariant $\theta(x, y \dots z)$; then will

$$\phi\left(\frac{d}{dx}, \frac{d}{dy} \dots \frac{d}{dz}\right) \cdot \theta(x, y \dots z)$$

be a contravariant of ϕ. For orthogonal transformations the terms contravariant and covariant coincide, and the theorem for this case appears to have been known to Mr Boole, see (15), same page. More generally, if ψ and θ be any two concomitants of ϕ, the algebraical product $\psi\theta$ will also be a concomitant of ϕ, provided that the systems of variables in ψ and θ have all distinct names, or that those which bear the same names are cogredient with one another. If this proviso does not hold good, the product in question will evidently be no longer a concomitant of ϕ. Let however Ψ denote what ψ becomes, and ϑ what θ becomes, when in place of the variables $x, y \dots z$ of every two contragredient synonymous systems in ψ and θ we write

$$\frac{d}{dx}, \frac{d}{dy} \dots \frac{d}{dz},$$

then will $\vartheta\psi$ and $\Psi\theta$ be each of them concomitants of ϕ, the synonymous systems becoming cogredient with ψ in the one case and with θ in the other.

(7) There is one principle of *paramount* importance which has not been touched upon in the preceding pages, which I am very far from supposing to exhaust the fundamental conceptions of the subject, (indeed, not to name other points of enquiry, I have reason to suppose that the idea of contragredience itself admits of indefinite extension through the medium of the reciprocal properties of commutants; the particular kind of contragredience hereinbefore considered having reference to the reciprocal properties of ordinary determinants only).

[9] *Camb. and Dub. Math. Journ.* Vol. vi. (1851), pp. 87–106.

[10] Mr Boole applied his theorem to obtain the cubic invariant of $(x, y)^4$, say $\phi(x, y)$, by operating upon its Hessian with $\phi\left(\frac{d}{dy}, -\frac{d}{dx}\right)$. More generally, when $\phi(x, y) = (x, y)^{2n}$, the catalecticant of the antepenultimate emanant of ϕ is also of the degree $2n$; and this, when operated upon by $\phi\left(\frac{d}{dy}, -\frac{d}{dx}\right)$, will give an invariant of the order $n + 1$, which is probably identical with the catalecticant of ϕ itself. There exists a most interesting transformation of the catalecticant of any emanant of a function of any degree in x, y, whether even or odd, under the form of a determinant some of the lines of which contain combinations only of x and y, without any of the coefficients, and all the rest the coefficients only of the given function without x or y. The Hessian being the catalecticant of the second emanant is of course included within this statement.

The principle now in question consists in introducing the idea of *continuous or infinitesimal* variation into the theory. To fix the ideas, suppose C to be a function of the coefficients of $\phi(x, y, z)$, such that it remains unaltered when x, y, z become respectively fx, gy, hz, provided that $fgh = 1$. Next, suppose that C does not alter when x becomes $x + \varepsilon y + \epsilon z$, when ε and ϵ are indefinitely small: it is easily and obviously demonstrable that if this be true for ε and ϵ indefinitely small, it must be true for *all* values of ε and ϵ. Again, suppose that C alters neither when x receives such an infinitesimal increment, y and z remaining constant, nor when y nor z separately receive corresponding increments, z, x and x, y in the respective cases remaining constant; it then follows from what has been stated above that this remains true for finite increments to x or y or z separately; and hence it may easily be shown that C will remain constant for any concurrent linear transformations of x, y, z, when the modulus is unity. This all-important principle enables us at once to fix the form of the symmetrical functions of the roots of $\phi\left(\frac{x}{y}, 1\right)$ which represent invariants of $\phi(x, y)$ when the coefficient of the p. 327

highest power of x is made unity. It also instantaneously gives the necessary and sufficient conditions to which an invariant of any given order of any homogeneous function whatever is subject, and thereby reduces the problem of discovering invariants to a definite form. But as these conditions coincide with those which have been stated to me as derived from other considerations by the gentleman whose labours in this department are concomitant with my own, I feel myself bound to abstain from pressing my conclusions until he has given his results to the press.

(8) By aid of the general principle enunciated in Note (6) above, we can easily obtain Aronhold's S and T. Let U be the given cubic function of x, y, z, and let $G(x, y, z; \xi, \eta, \zeta)$ be the polar reciprocal in respect to ξ, η, ζ of

$$\left(\xi \frac{d}{dx} + \eta \frac{d}{dy} + \zeta \frac{d}{dz}\right) U,$$

then $G(\xi, \eta, \zeta;\ x, y, z)$ as well as the former G will be a concomitant to U, but the homonymous systems of variables in the two G's will be contragredient; and, accordingly,

$$G\left(\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}; \frac{d}{d\xi}, \frac{d}{d\eta}, \frac{d}{d\zeta}\right) \cdot G(\xi, \eta, \zeta;\ x, y, z)$$

will be a concomitant to U; this concomitant is readily seen to be an invariant of the fourth order; that is, Aronhold's S. Again, from S, by means of the Eisenstein-Hermite theorem, we may derive a form $K(x, y, z)$ of the third degree in x, y, z, and whose coefficients will be of three dimensions; and, accordingly, if the Hessian of U be called $H(U)$,

$$K\left(\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}\right) \cdot H(U)$$

will be a Sextic Invariant of U, that is, Aronhold's T.